Gabriel Phun

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Summary— PhD in computational chemistry, now a solo dev building tools for AI agents, and with them. Built apps end to end, from mobile front ends to backend services and databases.

Qtpi Bonding LLC **Feb 2025–Present**

Solo Founder and Developer

Structmux | ●Rust, ●GPUI, ●Alacritty, Protocol Buffers, ACP (*built with AI coding agents*)

A tmux-style workspace; agent, terminal and markdown panes share a typed protobuf API over WebSocket and Zenoh.

- Lets users move and arrange nested-split panes from the keyboard, with directional focus; mouse optional.
- Lets users comment on note or agent passages and send them to an agent as one prompt; early, drafts temporary.
- Runs agents in panes with permission prompts and red/green edits; partly finished, some tools show raw input.

Gait | ●Rust, Protobuf, libgit2, Wasmtime (*built with AI coding agents*)

Coding agent harness where each turn is a fresh sandboxed process and state is folded from an append-only log.

- Records every session as an append-only log in its own git branch, so users can see what happened.
- Lets users restore code to any earlier run and rewind the transcript; snapshots never write to their repo.
- Asks users before any unallowed tool call, deny rules win, with a macOS-only kernel sandbox.

Archeglyph | ●TypeScript, Bun, ●SolidJS, Protobuf, elkjs (*built with AI coding agents*)

Renders deterministic SVG diagrams from node-and-edge text files, with a visual editor and PR-reviewable diffs.

- Renders byte-identical SVG for the same diagram, style and theme, and fails PR checks on stale SVGs.
- Diffs two diagram versions in place, with distinct colors and markers for added, deleted and modified parts.
- Posts one updated comment on same-repo PRs touching a diagram, with before and after if a base SVG exists.

Episutra | ●Rust, ●Flutter, PowerSync, ●Serverpod (*built with AI coding agents*)

Local-first knowledge-graph notes app with hybrid RAG search, CRDT collaboration, E2EE sync, and your own agent.

- Lets users search notes by exact words or by idea, with f: or s: limiting a query to one mode.
- Lets users run Claude Code, Codex or a local on-device model as a helper, with tools for notes and comments.
- Lets users set agent actions to allow, ask or deny, deletion denied by default, with edits as suggestions.

PocketCoder | ●Flutter, ●PocketBase, ●NixOS, Docker, ACP (*built with AI coding agents*)

Mobile app that provisions and runs your own self-hosted AI coding agent server, with no terminal needed.

- Lets the Pro app create a Linode server, boot PocketCoder on it and sign in without typing; Linode only.
- Lets users approve agent permission requests from their phone via chat or push, with unanswered ones expiring.
- Lets users choose coding agents, including Goose and Claude Code, on local Ollama or hosted providers.

Quanitya | ●Flutter, PowerSync, ●Serverpod (*built with AI coding agents*)

Track anything structured: end-to-end encrypted, customizable UI, and private JS scripts to analyze your data.

- Encrypts templates, entries, schedules and scripts on device before sync; server holds opaque encrypted text.
- Runs JavaScript analysis on device with statistics libraries, showing values or charts; no native desktop yet.
- Works locally by default with no server; sync is optional, via a self-hosted server or the Quanitya cloud.

Achaean | ●Flutter, ●Serverpod, ●PostgreSQL, ●Apache AGE, Git (*built with AI coding agents*)

Social network where mutual trust, not moderators, gates communities, and posts live in your own git repo.

- Grants community membership through mutual trust among README signers, above a threshold; still early.

Qtpi | ●DRF, ●Keycloak, ●Flutter, Matrix (*AI-assisted*)

Pay-per-match dating app designed to encourage intentional matching.

– Matches people using a web of trust that gauges closeness from stored connections and verifications.

SciPiPol | ●Django, ●PostgreSQL, HTMX, Docker (*AI-assisted*)

Platform that connects scientists to US state legislation, with ORCID sign-in and LegiScan bill data.

– Loads state legislation into Postgres, read by 21 unmanaged Django models that migrations never touch.

University of California, Riverside **Feb 2024–Feb 2025**

Postdoctoral Researcher

– 3 first-author publications

Skills

Languages ●Python, ●Dart, ●SQL, ●Bash, ●Django, ●DRF, ●Flutter, ●Bloc/Cubit, Drift, PowerSync, ●Serverpod

Infra / Tools Docker, docker-compose, ●PostgreSQL, PostGIS, SQLite, Git, Linux, GitHub Actions

APIs REST and RPC-style API design, Protobuf

Education

University of California, Irvine **2018–2023**

PhD, Computational Chemistry

– 3 first-author publications; DOE SCGSR Fellow, Los Alamos National Laboratory (2022–23)

Harvey Mudd College **2014–2018**

BS Chemistry + BS Physics

First/Co-First Author Publications

– Energy Transfer Mechanisms in Large Low-Bandgap Polymers from Time-Resolved Experiments and Nonadiabatic Molecular Dynamics Calculations - Phun, G. S. et al. *Chemistry of Materials* **2025**, *37*, 3769–3775

– Large-scale ab initio molecular dynamics for assessing stabilities of near-surface NV centers - Phun, G. S. et al. *Applied Physics Letters* **2025**, *126*, 171603

– Hydroselenation of olefins: elucidating the β-selenium effect - Phun, G. S. et al. *Chem. Sci.* **2024**, *15*, 20523–20533

– Comment on "Localized and Delocalized States of a Diamine Cation: Resolution of a Controversy" - Phun, G. S.; Wong, B. M. *J. Phys. Chem. Lett.* **2024**, *15*, 11415–11418

– Detailed-balance limits for sunlight-to-protonic energy conversion from aqueous photoacids and photobases based on reversible mass-action kinetics - Phun, G. S. et al. *Energy Environ. Sci.* **2023**, *16*, 4593–4611

– Constructing the Mechanism of Dinoflagellate Luciferin Bioluminescence Using Computation - Phun, G. S. et al. *J. Phys. Chem. Lett.* **2023**, *14*, 6001–6008

Coauthor Publications

– Majumdar, S. et al. *Journal of Computational Chemistry* **2025**, *46*, e70040

– Bhide, R. et al. *The Journal of Physical Chemistry A* **2024**, *128*, 4177–4188

– Franzke, Y. J. et al. *Journal of Chemical Theory and Computation* **2023**, *19*, 6859–6890

– Bhide, R. et al. *Journal of the American Chemical Society* **2022**, *144*, 14477–14488

– Kaur, M. et al. *The Journal of Physical Chemistry A* **2020**, *124*, 5474–5486

Patents

– Ardo, S. et al. Optically thin light-absorbers for increasing photochemical energy-conversion efficiencies, US Patent App. 16/673,680, 2020

– Ardo, S. et al. Membranes for enhancing rates of water dissociation and water formation, US Patent App. TBD, 2020